Problem Set 2

Instructor solutions

Euclid, continued fractions, and prime factorization

Week 2 | 100 points + 10 optional challenge points

Teaching note. Estimated working time is 2–3 hours, excluding the challenge. The continued fractions require only finite nesting and exact arithmetic. Subpart scores appear in parentheses. Award proof credit for the reasoning, including a valid alternative argument; avoid repeatedly penalizing a carried arithmetic error.

Instructions. Show your divisions and use exact fractions. Justify every proof or counterexample. You may use the division algorithm, Bézout's identity, and Euclid's lemma. You may use the fundamental theorem of arithmetic except when proving it in Problem 4.

Notation. A finite simple continued fraction is

$$[a_0; a_1, \dots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}, \qquad [a_0] = a_0.$$

Here a_0 is an integer and a_i is a positive integer for $i \geq 1$. Unless otherwise requested, use $a_n \geq 2$ when $n \geq 1$.

I. Euclid and continued fractions

1. One calculation, two readings

[14 points]

(a) Use the Euclidean algorithm to compute $\gcd(301, 133)$. (5)

(b) Work backward to find integers x, y such that $301x + 133y = \gcd(301, 133)$. Check your answer by substitution. (4)

(c) Use the quotients from (a) to express $301/133$ as a simple continued fraction. Reduce $301/133$ to lowest terms and explain why this does not change its continued-fraction expansion. (5)

Solution

(a) The divisions are

$$301 = 2 \cdot 133 + 35, \quad 133 = 3 \cdot 35 + 28,$$
$$35 = 1 \cdot 28 + 7, \qquad 28 = 4 \cdot 7 + 0.$$

Thus the gcd is 7.

(b) Reverse the divisions:

$$7 = 35 - 28 = 35 - (133 - 3 \cdot 35) = 4(301 - 2 \cdot 133) - 133 = 4 \cdot 301 - 9 \cdot 133.$$

Hence $(x, y) = (4, -9)$; indeed $1204 - 1197 = 7$.

(c) Dividing each line in (a) by its divisor gives the nested expression

$$\frac{301}{133} = 2 + \cfrac{1}{3 + \cfrac{1}{1 + \cfrac{1}{4}}} = [2; 3, 1, 4] = \frac{43}{19}.$$

Dividing every entry in the Euclidean divisions by 7 leaves the quotients $2, 3, 1, 4$ unchanged. Equivalently, the continued-fraction procedure acts on the value of the rational number, which is unchanged by reduction.

2. Unfolding a continued fraction [12 points]

(a) Evaluate $[2;3,4]$ and $[0;2,1,3]$ as fractions in lowest terms. (4)

(b) For $n \geq 1$ and $a_n \geq 2$, prove the identity

$$[a_0; a_1, \ldots, a_n] = [a_0; a_1, \ldots, a_{n-1}, a_n - 1, 1].$$

Explain the equality by looking at the end of the nested fraction. (4)

(c) Give a second expansion, ending in 1, for each number in (a). (4)

Solution

(a) Working from the inside outward,

$$[2;3,4] = 2 + \frac{1}{3 + 1/4} = 2 + \frac{4}{13} = \frac{30}{13}, \qquad [0;2,1,3] = \frac{1}{2 + 1/(1 + 1/3)} = \frac{1}{2 + 3/4} = \frac{4}{11}.$$

The displayed fractions are reduced: $30 = 2 \cdot 13 + 4$, $13 = 3 \cdot 4 + 1$, and $\gcd(4, 11) = 1$.

(b) The final tail has the same value because $[a_n - 1; 1] = (a_n - 1) + 1/1 = a_n$. Substituting equal tails into the same preceding nesting gives equal fractions. The assumption $a_n \geq 2$ ensures that the new entry $a_n - 1$ is positive.

(c) The second expansions are $30/13 = [2;3,3,1]$ and $4/11 = [0;2,1,2,1]$.

3. Bounding a square root with fractions [12 points]

Consider the four truncations

$$c_0 = [1], \qquad c_1 = [1;2], \qquad c_2 = [1;2,2], \qquad c_3 = [1;2,2,2].$$

Such truncations are called *convergents*.

(a) Compute the four fractions in lowest terms. (4)

(b) Decide which are below $\sqrt{2}$ and which are above it by comparing their squares with 2. Do not use decimal approximations. (4)

(c) Using these values, give the narrowest interval you can with a lower endpoint below $\sqrt{2}$ and an upper endpoint above it. Find its exact width. (4)

Solution

(a) Direct nesting gives $c_0 = 1$, $c_1 = 3/2$, $c_2 = 7/5$, and $c_3 = 17/12$.

(b) For a positive fraction p/q, the sign of $(p/q)^2 - 2$ is the sign of $p^2 - 2q^2$:

Truncation	p/q	$p^2 - 2q^2$	Comparison
c_0	$1/1$	-1	$c_0 < \sqrt{2}$
c_1	$3/2$	1	$c_1 > \sqrt{2}$
c_2	$7/5$	-1	$c_2 < \sqrt{2}$
c_3	$17/12$	1	$c_3 > \sqrt{2}$

Squaring preserves inequalities between positive numbers, so these comparisons prove the asserted bounds.

(c) The larger lower bound is $7/5$, and the smaller upper bound is $17/12$. Hence

$$\frac{7}{5} < \sqrt{2} < \frac{17}{12}, \qquad \frac{17}{12} - \frac{7}{5} = \frac{85 - 84}{60} = \frac{1}{60}.$$

II. Unique factorization and its consequences

4. Why factorization exists and is unique [14 points]

Prove the fundamental theorem of arithmetic in the following steps.

(a) Use strong induction to prove that every integer $n > 1$ is a product of primes. Treat the prime and composite cases separately. (6)

(b) Suppose $p_1 \cdots p_r = q_1 \cdots q_s$, where all factors are prime. Use Euclid's lemma to match and cancel prime factors. Explain why this process proves uniqueness up to order, including why one side cannot have factors left over. (6)

(c) Why would allowing 1 as a prime destroy this uniqueness statement? (2)

Solution

(a) The base case $n = 2$ is prime. Assume all integers m with $2 \le m < n$ are products of primes. If n is prime, it already is such a product. If n is composite, then $n = ab$ for integers $1 < a, b < n$. By the inductive hypothesis, each of a, b is a product of primes; multiplying those products gives a prime factorization of n.

(b) Since p_1 divides $q_1 \cdots q_s$, repeated use of Euclid's lemma shows that $p_1 \mid q_j$ for some j. Because q_j is prime and $p_1 > 1$, it follows that $p_1 = q_j$. Reorder the q's and cancel this factor. Repeat the same argument with the shorter products. The number of factors decreases at every step, so the process terminates. If only one side had remaining factors, their product would be greater than 1, while the empty product on the other side would equal 1. This is impossible. Thus $r = s$ and the factors agree up to order.

(c) Inserting arbitrarily many 1's would give, for example, $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3$. These lists differ in length and cannot be obtained from each other by reordering.

5. Completing a square or a cube [14 points]

Let $N = 5400$.

(a) Find the prime factorization of N. (4)

(b) Find the least positive integer k for which Nk is a perfect square. Give the square root of Nk and prove that your k is least. (5)

(c) Find the least positive integer ℓ for which $N\ell$ is a perfect cube. Give the cube root of $N\ell$ and prove minimality. (5)

Solution

(a) $5400 = 54 \cdot 100 = 2^3 \cdot 3^3 \cdot 5^2$.

(b) A square has an even exponent at every prime. The exponents of 2 and 3 in N are odd, so any working multiplier must contain at least one factor of 2 and one of 3. Thus $6 \mid k$ and $k \ge 6$. Taking $k = 6$ works:

$$5400 \cdot 6 = 2^4 \cdot 3^4 \cdot 5^2 = (2^2 \cdot 3^2 \cdot 5)^2 = 180^2 = 32400.$$

(c) In a cube, every prime exponent is divisible by 3. The exponents of 2 and 3 already satisfy this; the exponent 2 of 5 needs at least one additional factor of 5. Therefore every working ℓ is divisible by 5 and is at least 5. This is attained:

$$5400 \cdot 5 = 2^3 \cdot 3^3 \cdot 5^3 = 30^3 = 27000.$$

The exponent tests in both parts follow from uniqueness of prime factorization, since taking a square or cube doubles or triples each exponent.

6. True, false, or missing a hypothesis? [12 points]

All variables below are positive integers.

(a) Prove or disprove: if $a^2 \mid b^2$, then $a \mid b$. (3)

(b) Prove or disprove: if $a \mid b^2$, then $a \mid b$. (3)

(c) Give a counterexample to the claim that $a \mid c$ and $b \mid c$ imply $ab \mid c$. Then prove that the claim becomes true when $\gcd(a, b) = 1$. (6)

Solution

(a) True. For any prime p, let α be its exponent in a and β its exponent in b, allowing zero exponents. Divisibility $a^2 \mid b^2$ gives $2\alpha \leq 2\beta$, hence $\alpha \leq \beta$. This holds for every prime, so $a \mid b$ by unique factorization.

(b) False. Take $a = 4$ and $b = 2$: $4 \mid 2^2$, but $4 \nmid 2$. A prime dividing a square must divide its base, but a composite divisor can require more copies of a prime than the base contains.

(c) Take $a = 4$, $b = 6$, and $c = 12$. Both 4 and 6 divide 12, whereas 24 does not. Now assume $\gcd(a, b) = 1$. No prime occurs in both a and b. At any prime, its exponent in ab is therefore its exponent in one of the factors (or zero). Since each factor divides c, that exponent is at most the exponent in c. Hence $ab \mid c$. Alternatively, write $c = au$ and use Bézout for (a, b) to deduce $b \mid u$, yielding $c = abv$.

III. Finding primes and knowing when to stop

7. The sieve of Eratosthenes [12 points]

(a) Use the grid to sieve the integers from 2 through 120. Mark 1 as neither prime nor composite. Process primes in increasing order, crossing out their multiples beginning at their squares. List all primes that remain and give their total number. (8)

(b) The numbers 91 and 119 survive the passes for $2, 3, 5$. Explain why they are nevertheless composite and identify the pass that removes them. (4)

Solution

(a) Since $10^2 < 120 < 11^2$, the required sieving primes are $2, 3, 5, 7$. Starting at $4, 9, 25, 49$, respectively, and crossing out multiples leaves the following 30 primes:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29,$$
$$31, 37, 41, 43, 47, 53, 59, 61, 67, 71,$$
$$73, 79, 83, 89, 97, 101, 103, 107, 109, 113.$$

For checking a student's grid, the numbers of composites first removed in the four passes are $59, 19, 7, 4$, totaling 89. Thus $1 + 89 + 30 = 120$, including the separate entry for 1.

(b) $91 = 7 \cdot 13$ and $119 = 7 \cdot 17$. Both are removed by the pass for 7. Failure to have a factor among $2, 3, 5$ does not rule out a factor of 7.

8. Why the sieve works [10 points]

(a) Prove that every composite integer $n > 1$ has a prime divisor at most $\sqrt{n}$. Deduce that sieving by all primes $p \le \sqrt{N}$ removes every composite integer at most N. (4)

(b) Explain why the pass for a prime p may begin at p^2: every multiple mp with $2 \le m < p$ has already been removed. Your argument must also cover composite m. (4)

(c) Would the sieve always work if we used only primes $p < \sqrt{N}$? Give a counterexample showing why the equality matters. (2)

Solution

(a) Write $n = ab$ with $1 < a \le b$. If $a > \sqrt{n}$, then also $b > \sqrt{n}$, so $ab > n$, a contradiction. Therefore $a \le \sqrt{n}$. Existence of prime factorization gives a prime divisor p of a; then $p \mid n$ and $p \le a \le \sqrt{n}$. For $n \le N$ this yields $p \le \sqrt{N}$, so a processed prime removes n. No prime is removed, because every crossed-out number has a proper prime divisor.

(b) Every integer $m \ge 2$ has a prime divisor $q \le m$. When $m < p$, this gives $q < p$. Thus $q \mid mp$, and the earlier pass for q already removed mp. Even when the earlier pass began at q^2, it reaches mp because $m \ge q$ and $p > q$ give $mp > q^2$. This reasoning does not assume that m itself is prime.

(c) Take $N = 49$. The primes strictly below $\sqrt{49} = 7$ are $2, 3, 5$; none divides 49. The composite $49 = 7^2$ survives unless the pass for 7 is included. More generally, $N = p^2$ for any prime p illustrates the same issue.

Optional challenge: a square equals a cube [10 extra points]

Let a, b be positive integers satisfying $a^2 = b^3$. Prove that there is a positive integer t such that $a = t^3$ and $b = t^2$. *Hint:* compare the exponents of a fixed prime on the two sides.

Solution

Let the exponents of a prime p in a and b be α_p and β_p. Uniqueness of prime factorization gives $2\alpha_p = 3\beta_p$. Since $3 \mid 2\alpha_p$ and $3 \nmid 2$, Euclid's lemma gives $3 \mid \alpha_p$. Write $\alpha_p = 3\gamma_p$; the equality then gives $\beta_p = 2\gamma_p$. Set $t = \prod_p p^{\gamma_p}$, a finite product. Its cube is a and its square is b. If $a = b = 1$, take $t = 1$ (the empty product). Conversely, any $a = t^3$, $b = t^2$ satisfies $a^2 = b^3 = t^6$.
Suggested credit: exponent equation (3), divisibility of the exponents (3), construction and verification of t (4).